



UG Program in Artificial Intelligence & Data Science

Experiment No. – 5				
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Program Execution/ formation/ correction/ ethical practices (06)	Professional Ethics (02)	Viva (02)	Experiment Total (10)	Sign with Date

Experiment No. 5

5.1 Aim: To demonstrate the process of creating a Hive database, loading data, and conducting descriptive analytics on a dataset.

5.2 Course Outcome: Collect, manage, store, query and analyze various forms of Big Data.

5.3 Learning Objectives: To introduce programming skills to build simple solutions using big data technologies such as MapReduce and scripting for NoSQL, and the ability to write parallel algorithms for multiprocessor execution.

5.4 Requirement: Hive database

5.5 Related Theory:

Hive: Apache Hive is a distributed, fault-tolerant data warehouse system that enables analytics at a massive scale. Hive Metastore(HMS) provides a central repository of metadata that can easily be analyzed to make informed, data driven decisions, and therefore it is a critical component of many data lake architectures. Hive is built on top of Apache Hadoop and supports storage on S3, adls, gs etc though hdfs. Hive allows users to read, write, and manage petabytes of data using SQL.

Create Database Statement:

- Create Database is a statement used to create a database in Hive. A database in Hive is a **namespace** or a collection of tables. The **syntax** for this statement is as follows:

```
CREATE DATABASE|SCHEMA [IF NOT EXISTS] <database name>
```

- Here, IF NOT EXISTS is an optional clause, which notifies the user that a database with the same name already exists. We can use SCHEMA in place of DATABASE in this command. The following query is executed to create a database named **userdb**:



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```
hive> CREATE DATABASE [IF NOT EXISTS] userdb;
```

or

```
hive> CREATE SCHEMA userdb;
```

- The following query is used to verify a databases list:

```
hive> SHOW DATABASES ;  
default  
userdb
```

Features of Hive

- These are the following features of Hive:
- Hive is fast and scalable.
- It provides SQL-like queries (i.e., HQL) that are implicitly transformed to MapReduce or Spark jobs.
- It is capable of analyzing large datasets stored in HDFS.
- It allows different storage types such as plain text, RCFile, and HBase.
- It uses indexing to accelerate queries.
- It can operate on compressed data stored in the Hadoop ecosystem.
- It supports user-defined functions (UDFs) where user can provide its functionality

5.6 Procedure:

(NOTE: U can consider any database)

Commands: -

1. First download the .csv file(Pseudo_facebook.csv) from the drive link ([pseudo_facebook](#))
2. Add this csv file in one folder on the main operating system (Here FB_Data).
3. Mount this folder in the Cloudera. This can be done as:
 - Cloudera->Devices->Shared Folder->Add new->Give Folder location->Check auto mount and make permanent->Click ok.
 - Open Terminal->Type su and enter password (cloudera) to enter into root directory ->Create directory using mkdir command->Mount the folder in created directory using the command 'mount -t vboxsf folder_name /home/cloudera/directory_name/'.
4. Open cloudera terminal and type hive to use hive.
5. To Show databases ***show databases;***
6. To create a database and use it.

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create database trial; use trial;

7. To create table.

```
create table facebook (id int,age int,day int,year int month int,gender
string,tенure int,friends int,friend_init int,likes int,likes_recd int,mlikes
int,mlikes_recd int,wlikes int,wlikes_recd int)
```

>row format

delimited >fields

terminated by ','

8. To load the data. *Load data local inpath*

'/home/cloudera/directory_name/Pseudo_facebook.csv'
into table facebook;

9. To check whether the data is loaded or not.

Open browser open **Hue** in that open file browser and inside it open
/user/hive/warehouse/trial.db/facebook/Pseudo_facebook
k.csv

10. Execute the following queries:

```
query1:find out the total no of users in the dataset.
query2:find out the no of fb users above the age of 25.
query3:male fb users tend to have more female users.
query4:how many likes do young people receive on fb oppose to old people.
query5:find out the count of fb users for birthday month.
query6:do young members use mobile or computers for fb browsing.
```

11. To find the total number of users in the dataset.

```
select count(*) from facebook;
```

12. To find the number of facebook users above the age of 25.

select count() from facebook where age>=25;*

13. Male facebook user tend to have more female users.

```
select avg(friends) from facebook where gender='male';
```

```
select avg(friends) from facebook where gender='female';
```

```
select gender,avg(friends) from facebook group by gender.
```

14. How many likes do young people receive on facebook oppose to old people?

```
select avg(likes_recd) from facebook where age>=13 and
age<=25; select avg(likes_recd) from facebook where
age>=35;
```

15. Find out the count of facebook users for birthday month.

Select month,count() from facebook group by month;*

16. Do young members use mobile or computers for facebook browsing?

*select avg(mlikes),avg(wlikes) from facebook where age>=13 and
age<=25;*

5.7 Program and Output:



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```
File Edit View Search Terminal Help
[cloudera@quickstart ~]$ su
Password:
[root@quickstart cloudera]# mkdir fb
mkdir: cannot create directory 'fb': File exists
[root@quickstart cloudera]# mount -t vboxsf FB_data /home/cloudera/fb/
[root@quickstart cloudera]#
```



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```
[cloudera@quickstart ~]$ hive

Logging initialized using configuration in file:/etc/hive/conf.d/hive-log4j.p
roperties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.
hive> show databases;
OK
default
trial
Time taken: 0.466 seconds, Fetched: 2 row(s)
hive> create database facebook;
OK
Time taken: 1.124 seconds
hive> use facebook;
OK
Time taken: 0.06 seconds
hive> create table info (id int,age int,day int,year int,month int,gender strin
g,t tenure int,friends int,friend_init int,likes int,likes_rec int,mli
kes_rec int,wlikes int,wlikes_rec int)
> row format
> fields
> ,
> row format delimited >fields terminated by ','
> row format delimited >fields terminated by ',' >stored as textfile;
Nov14/2018 12:28:11
at org.apache.hadoop.hive.ql.parse.HiveParser.tableRowFormat(HiveParser.java:34687)
at org.apache.hadoop.hive.ql.parse.HiveParser.createTableStatement(HiveParser.java:5196)
at org.apache.hadoop.hive.ql.parse.HiveParser.delStatement(HiveParser.java:2557)
at org.apache.hadoop.hive.ql.parse.HiveParser.execStatement(HiveParser.java:1589)
at org.apache.hadoop.hive.ql.parse.HiveParser.statement(HiveParser.java:1605)
at org.apache.hadoop.hive.ql.parse.ParseDriver.parse(ParseDriver.java:201)
at org.apache.hadoop.hive.ql.parse.ParseDriver.parse(ParseDriver.java:166)
at org.apache.hadoop.hive.ql.Driver.compileDriver.java:522)
at org.apache.hadoop.hive.ql.Driver.compileInternal(Driver.java:1356)
at org.apache.hadoop.hive.ql.Driver.runInternal(Driver.java:1473)
at org.apache.hadoop.hive.ql.Driver.run(Driver.java:1285)
at org.apache.hadoop.hive.ql.Driver.run(Driver.java:1275)
at org.apache.hadoop.hive.cli.CliDriver.processLocalCmd(CliDriver.java:226)
at org.apache.hadoop.hive.cli.CliDriver.processCmd(CliDriver.java:175)
at org.apache.hadoop.hive.cli.CliDriver.processLine(CliDriver.java:389)
at org.apache.hadoop.hive.cli.CliDriver.executeDriver(CliDriver.java:781)
at org.apache.hadoop.hive.cli.CliDriver.run(CliDriver.java:699)
at org.apache.hadoop.hive.cli.CliDriver.main(CliDriver.java:634)
at sun.reflect.NativeMethodAccessorImpl.invoke(Native Method)
at sun.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:57)
at sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
at java.lang.reflect.Method.invoke(Method.java:606)
at org.apache.hadoop.util.RunJar.run(RunJar.java:221)
at org.apache.hadoop.util.RunJar.main(RunJar.java:136)
FAILED: ParseException line 2:0 cannot recognize input near 'row' 'format' 'fields' in table row format specification

hive> create table info (id int,age int,day int,year int,month int,gender string,tenure int,friends int,friend_init int,likes int,likes_rec int,mli
kes_rec int,wlikes int,wlikes_rec int)
> row format delimited >fields terminated by ',' >stored as textfile;
Nov14/2018 12:30:10: ( tableRowFormatMapKeyIdentifier ???)
at org.apache.hadoop.hive.ql.parse.HiveParser.tableRowFormat(HiveParser.java:34687)
at org.apache.hadoop.hive.ql.parse.HiveParser.createTableStatement(HiveParser.java:5196)
at org.apache.hadoop.hive.ql.parse.HiveParser.delStatement(HiveParser.java:2557)
at org.apache.hadoop.hive.ql.parse.HiveParser.execStatement(HiveParser.java:1589)
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at org.apache.hadoop.hive.ql.Driver.compileDriver.java:522)
at org.apache.hadoop.hive.ql.Driver.compileInternal(Driver.java:1356)
at org.apache.hadoop.hive.ql.Driver.runInternal(Driver.java:1473)
at org.apache.hadoop.hive.ql.Driver.run(Driver.java:1285)
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at org.apache.hadoop.hive.cli.CliDriver.processLine(CliDriver.java:389)
at org.apache.hadoop.hive.cli.CliDriver.executeDriver(CliDriver.java:781)
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at sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
at java.lang.reflect.Method.invoke(Method.java:606)
at org.apache.hadoop.util.RunJar.run(RunJar.java:221)
at org.apache.hadoop.util.RunJar.main(RunJar.java:136)
FAILED: ParseException line 2:21 cannot recognize input near '>' 'fields' 'terminated' in serde properties specification
```

5.8 Conclusion:

This experiment successfully demonstrated the process of creating a Hive database, loading data, and performing descriptive analytics on a social media dataset. We utilized various Hive Query Language (HQL) commands to create tables, load external CSV data, and execute analytical queries. The practical application of these commands allowed us to extract meaningful insights, such as the total number of users, the distribution of users by age and gender, and engagement metrics like likes received and browsing device preferences. This exercise effectively showcased Hive's capabilities as a powerful data warehousing system for managing and analyzing large datasets within the Hadoop ecosystem, thereby achieving the stated aim and learning objectives.

5.9 Questions:

4 to 5 MCQ/Fill in the Blanks

1. Which command is used to load data from a file into a Hive table?

- A) LOAD DATA INPATH 'file_path' INTO TABLE table_name;
- B) INSERT INTO table_name VALUES ('file_path');
- C) ADD DATA 'file_path' TO table_name;
- D) IMPORT DATA 'file_path' INTO table_name;



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Answer: A) LOAD DATA INPATH 'file_path' INTO TABLE table_name;

2. Which Hive command is used to retrieve the first 10 records from a table?

- A) SELECT * FROM table_name LIMIT 10;
- B) FETCH FIRST 10 ROWS FROM table_name;
- C) GET 10 FROM table_name;
- D) SHOW 10 RECORDS IN table_name;

Answer: A) SELECT * FROM table_name LIMIT 10;

3. What is the output of the command SELECT COUNT(*) FROM table_name;?

- A) The sum of all numeric values in the table
- B) The total number of records in the table
- C) The number of columns in the table
- D) The first record of the table

Answer: B) The total number of records in the table

4. What is the purpose of the HAVING clause in a Hive query?

- A) To filter records before grouping
- B) To filter records after grouping
- C) To limit the number of records returned
- D) To join multiple tables

Answer: B) To filter records after grouping

5. What function would you use to calculate the average of a column in Hive?

- A) AVG(column_name);
- B) MEAN(column_name);
- C) SUM(column_name) / COUNT(column_name);
- D) Both A and C

Answer: D) Both A and C